



stone paper scissors!

The Game

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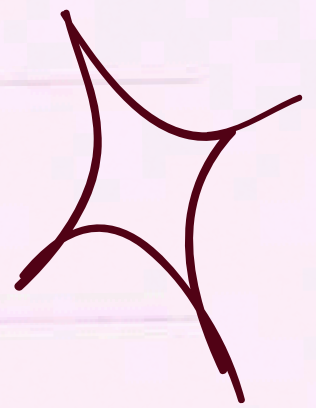


Ideation

Have you ever wanted to play stone, paper and scissors, but never found a friend to play with? Our game is inspired by the classic game of stone, paper and scissors, except you can play against a computer! So instead of finding real human friends, you can play by yourself against a computer!



components



- 3 green LEDs: indicates the player's move
- 3 red LEDs: indicates the player's move
- 3 push buttons: for taking the player's move as an input
- Buzzer: to indicate the start of the game and make different sounds at the end of the game
- Neopixel: to show the winner of each game (blue for when user win, red for when computer wins and orange for a tie)
- ESP32: to connect the circuit to the computer
- Power supply: to supply 5 volts to the neopixel
- Breadboard, Jumper/Copper wires to make the circuit

player guide aka logic

Step 1: Choose your move and press-and-hold its button to lock it in and start the game

Step 2: When the timer ends, the computer's move is revealed through the corresponding LED

Step 3: After comparing both moves the results are revealed using light and sound.

- **If the player wins:** NeoPixel glows blue & buzzer plays long melodic tone
- **If the computer wins:** NeoPixel glows red & buzzer plays two short beeps
- **If they tie:** NeoPixel glows orange & buzzer plays two short beeps

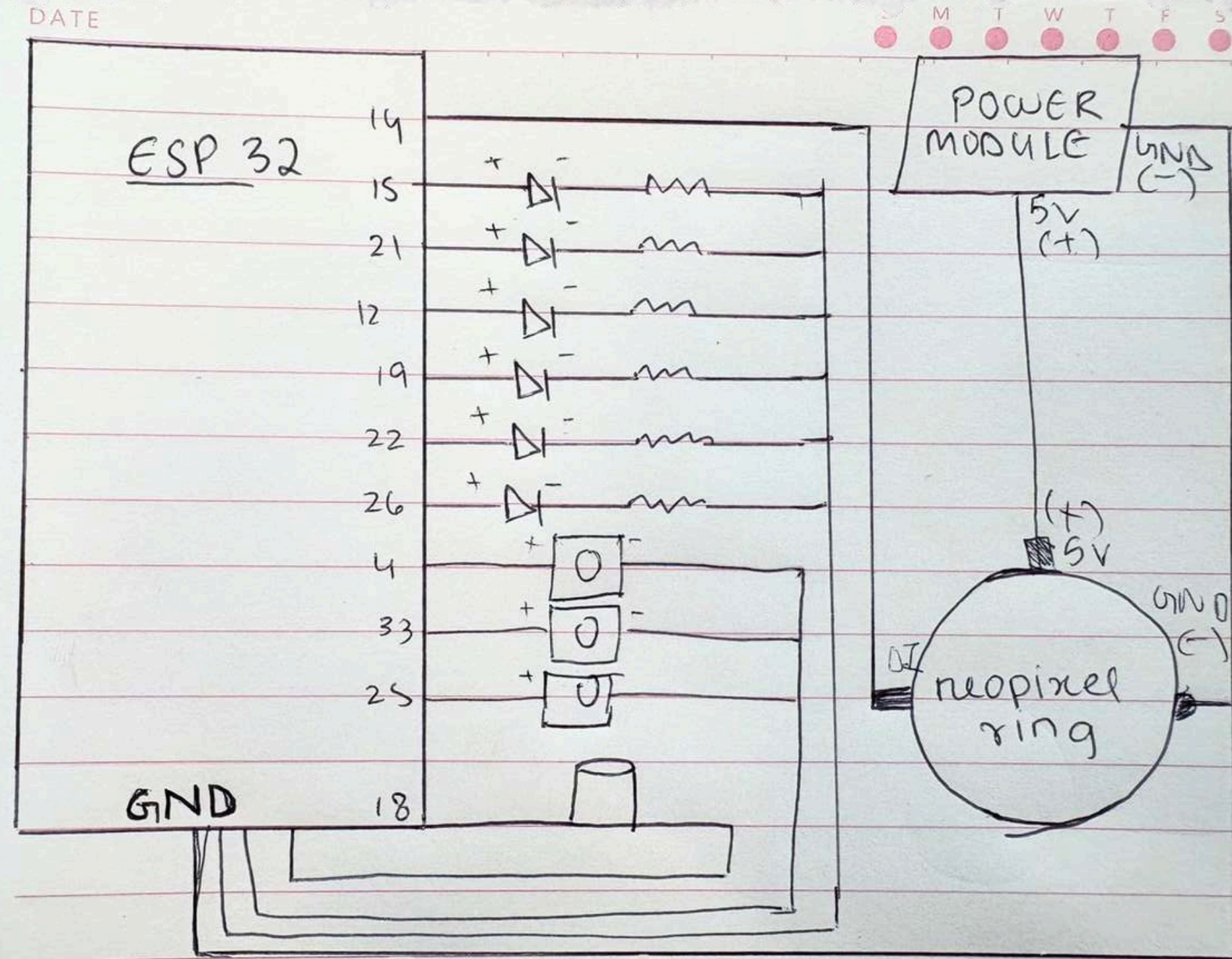


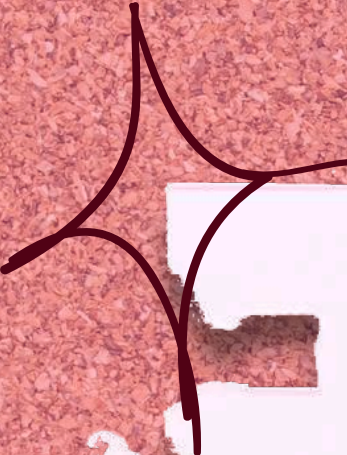
Flow



- Importing all the modules
- Assigning pins and variable names to each component
- Making a list with all the moves and LED names which corresponds to the computer's moves
- Defining necessary functions to calculate the winner and a timer
- Initializing the game condition
- Conditions for all the three possible moves the player can play

circuit diagram



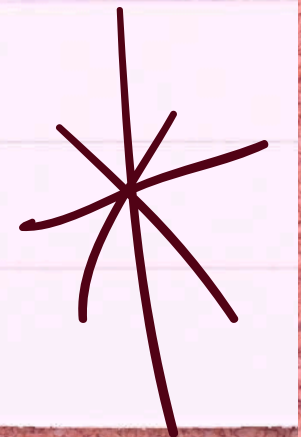


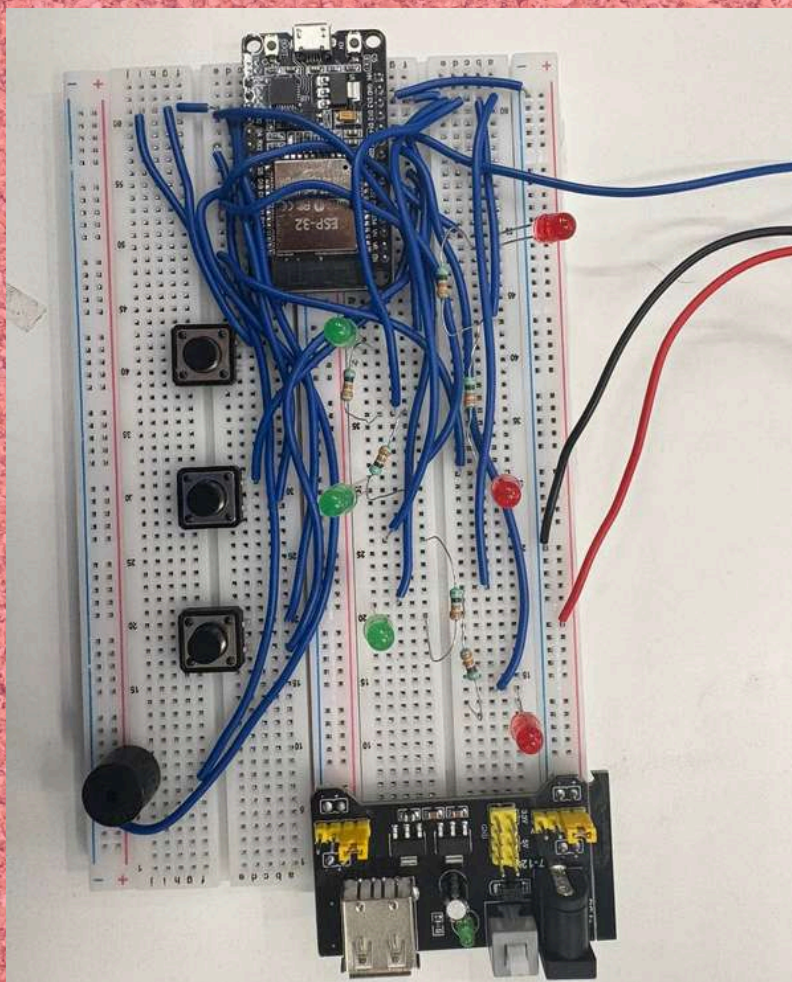
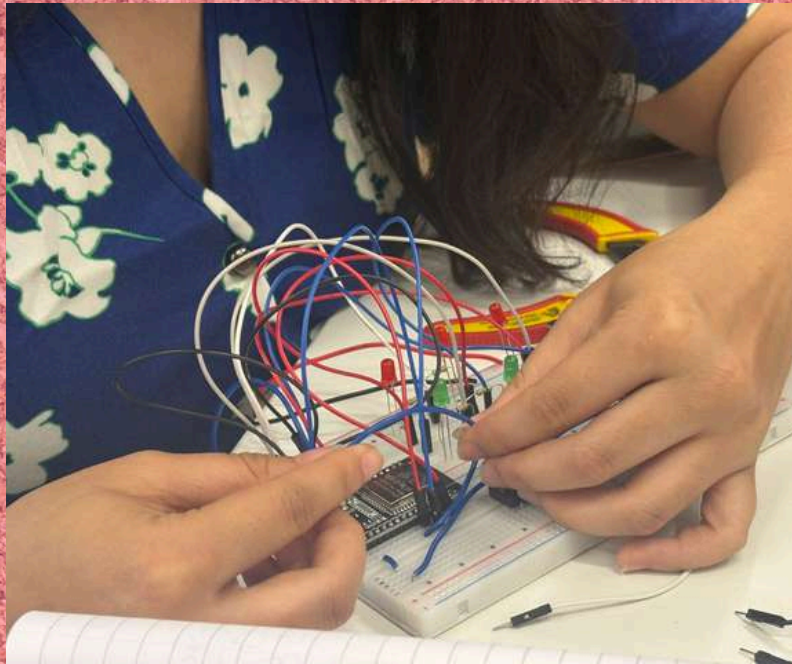
learnings

- We learned how to debug when there are multiple different components in a single circuit
- We learnt how use basic logic and simple flow to make something interesting
- We learnt that once you understand the logic of the code its easy to use the right commands to get the outcome you need

pain points

- How to make the buzzer and the neopixel **operate simultaneously** while using `time.sleep()`
- How to update the user's input without stopping the for loop
- Keeping track of **all components** and if they work without any bugs
- Keeping the wire **setup minimal** without using too many wires and connection
- How to make it **engaging** and **interestiing** without over complicating the code





Individual contribution

1. Ideation process of the concept
2. Input in discussions for the main code
3. Import and initialization codes
4. Entire circuit formation and attachment on the breadboard
 - Rough circuit formation using jumper wires and then the final circuit using copper wires.

Thanks!

